



DESCRIPTION OF COURSEWORK

Course Code	SWE311
Course Name	Mobile Application System Design
Lecturer	Nurul Amera binti Muhamad Nazmi
Academic Session	2025/09
Assessment Title	Assignment 2

A. Introduction/ Situation/ Background Information

This assignment will evaluate students' ability to apply mobile app development concepts by designing and implementing a simple CRUD (Create, Read, Update, Delete) feature. Students may use SQLite, Firebase, or SharedPreferences to store and manage data. This assignment helps students understand how data is handled, displayed, and updated in a mobile app.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

- CLO 1 Demonstrate ability to apply the mobile application architecture.
- CLO 2 Evaluate navigation schemes in mobile application.
- CLO 3 Analyze related programming languages to develop basic features in mobile application.
- CLO 4 Display an awareness for continuous learning and analysis of mobile application technology.

D. Instruction to Students

This is an **individual assignment**. You are required to submit a comprehensive report (1,500–2,000 words, PDF format) containing the following sections: App Introduction, CRUD Implementation Explanation, Screenshots of Functional App, and Reflection. Alongside the report, you must include the **working app file or APK (zipped)**.

Ensure that the files follow the proper naming convention: **StudentID_Assignment2_Report.pdf** for the report and **StudentID_Assignment2_App.zip** for the application. All files must be uploaded to the Moodle by the specified due date

The deadline for this assignment is on **5 December 2025, 11:59 pm**. No submission will result in 0 marks for your Assignment. All submissions shall be made via the Moodle platform.

E. Evaluation Breakdown

No.	Component Title	Mark
1.	App Concept & Design Consistency (CLO1)	10
2.	CRUD Feature Functionality (CLO2)	10
3.	Code Explanation & Reflection (CLO3, CLO4)	10
	TOTAL	30

F. Task(s)

Each student must design and develop a basic mobile application that performs CRUD operations on simple data (e.g., notes, contacts, or tasks). The app should focus on functionality, usability, and correctness rather than complex design.

Task 1: Use current mobile application design for development (CLO1)

- a. Select a real-world app idea and apply consistent layouts (color, font, and structure) for all screens.
- b. Introduce your app concept and explain its relevance to daily use.
- c. Design at least 4 screens that represent your app structure using **Flutter framework**:
 - I. Login/Sign-up Page
 - II. Home / Dashboard
 - III. Feature / Content Page
 - IV. Profile / Settings Page

Task 2: Apply knowledge and practical skills to design basic features (CLO2)

Implement one full CRUD functionality using either:

- a. SQLite (local database)
- b. Firebase (cloud database)
- c. SharedPreferences (simple key-value storage)

Ensure users can **add, view, edit, and delete** records.

Task 3: Analyze related programming concepts for app design (CLO3)

- a) In your report, briefly explain how each CRUD function works (Insert, Update, Delete, Retrieve).
- b) Include short code snippets with explanations of what each part does (no need for full code).

Task 4: Display awareness for continuous learning and design reflection (CLO4)

- a) Reflect (150–200 words) on what you learned while developing the CRUD feature for example, what challenges you faced, how you solved them, and what you would improve next time.